

Zabbix Email Alert Configuration (Media Type)

Goal: Configure Zabbix from scratch to monitor a Linux host and receive Problem and Recovery email notifications.

Architecture

Linux Host (Zabbix Agent)



Zabbix Server



Trigger



Action



User Media



SMTP Email



Inbox

Prerequisites

1. Zabbix Server installed
2. Zabbix Frontend accessible

3. Ubuntu/Linux host
4. Zabbix Agent installed
5. SMTP server details
6. Email account for notifications

Overall Workflow

1. A[Create Host Group]
2. B[Add Host]
3. C[Install & Configure Agent]
4. D[Link Template]
5. E[Verify Latest Data]
6. F[Create Trigger]
7. G[Configure Email Media Type]
8. H[Create User Group]
9. I[Create User]
10. J[Add Email Media]
11. K[Create Action]
12. L[Test CPU Load]
13. M[Problem Generated]
14. N[Email Received]
15. O[Recovery Email]

Step 1 - Create a Host Group

A Host Group in Zabbix is a logical container used to organize hosts (servers, VMs, network devices, applications, etc.). It does not monitor anything by itself; instead, it helps you manage and control multiple hosts efficiently.

Navigation

```
Configuration
└─ Host groups
```

— Create host group

Fill the following fields

Field Value

Host group Name= Linux Servers/Windows Server/(custom)

Click **Add**

Shown in SS1,02

The screenshot displays the Zabbix web interface. On the left is a sidebar with navigation menus: Monitoring, Services, Inventory, Reports, Configuration (selected), Administration, Support, Integrations, and Help. Under 'Configuration', 'Host groups' is selected. The main panel shows the 'Host groups' list with columns for Name, Hosts, Templates, Members, and Info. A 'Create host group' button is in the top right. Below the list, a modal dialog is open for creating a new group. The 'Group name' field contains 'rhel' and the 'Add' button is highlighted. The URL at the bottom of the browser window is 'http://192.168.44.128/zabbix/templates.php?filter_set=1&filter_groups[0]=5'. The footer of the Zabbix interface reads 'Zabbix 6.0.46, © 2001–2026, Zabbix SIA'.

Step 2 - Add a Host

Navigation



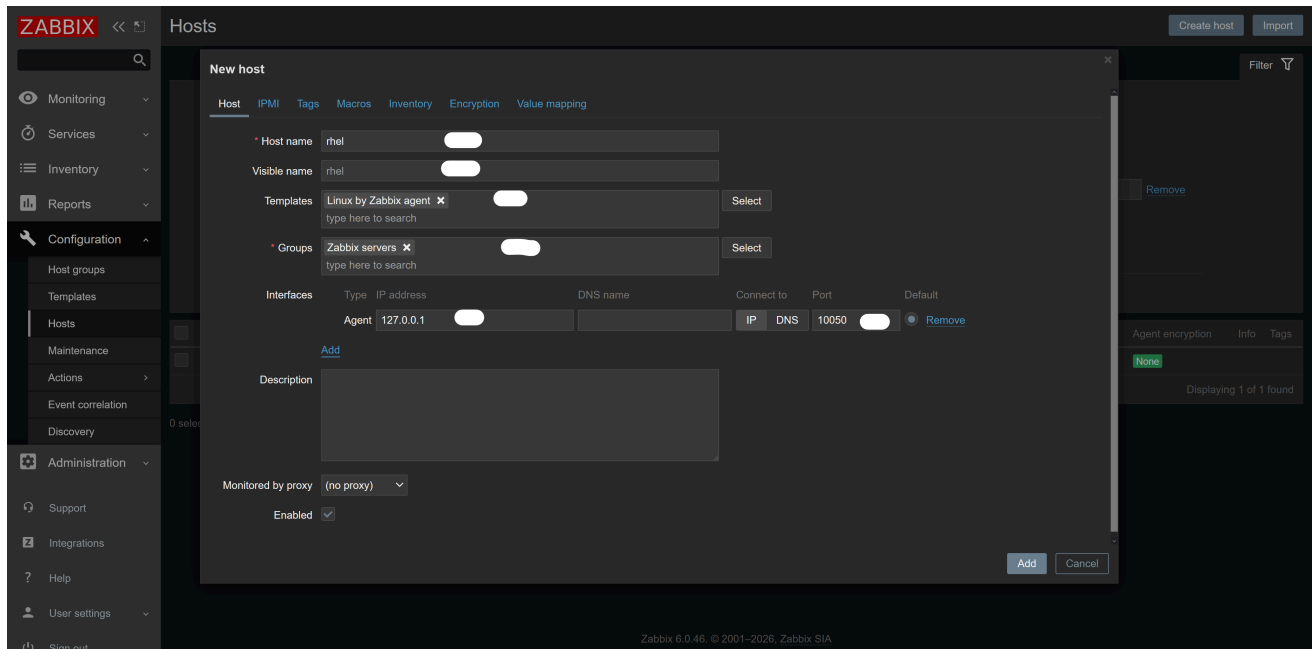
Field	Example
Host name	wsrv-445 , lserver229 / <i>(Custom)</i>
Visible name	wsrv-445 / <i>(Custom)</i>
Host groups	Linux Servers , Windows Servers , <i>(Custom)</i> (multiple groups allowed)
Templates	Linux by Zabbix Agent , Windows by Zabbix Agent

On Interfaces Click Add

Field	Value
Type	Agent
Connect to	IP
IP Address	192.168.44.130
Port	10050

Click **Add**

Shown in SS2



Step 3 – Add Trigger

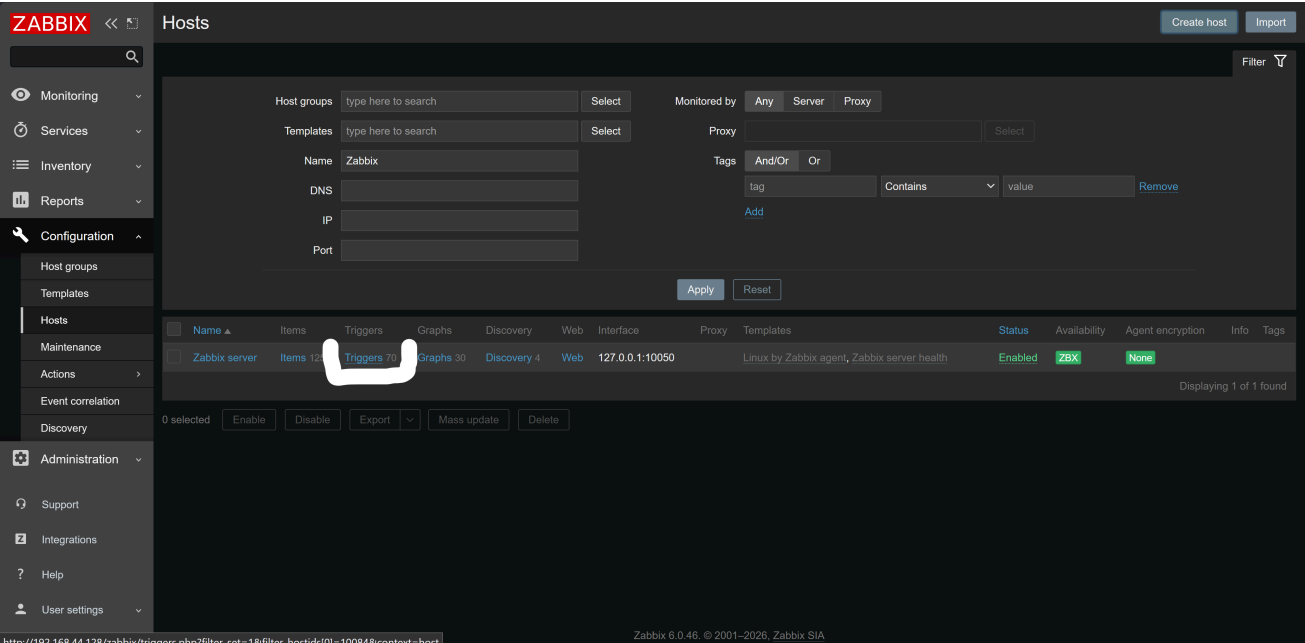
A trigger checks the collected data and decides whether the host is in a normal (OK) state or a problem (PROBLEM) state.

Navigation

```
Configuration
├── Hosts
│   └── Triggers
│       └── Create trigger
```

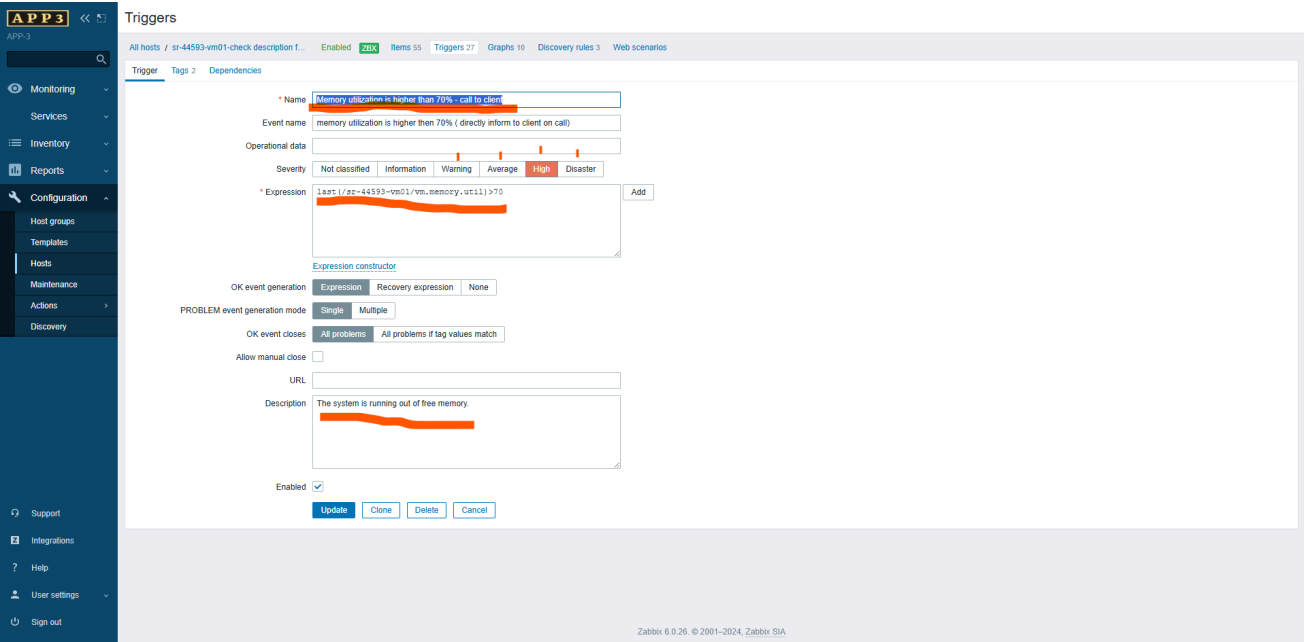
[OR YOU CAN CLONE EXISTING TRIGGER]

Shown in SS3



Trigger Tab

Shown in SS4



Field	Value
Name	Memory utilization is higher than 70% - call to client
Event name	memory utilization is higher than 70% (directly inform to client on call)

Field	Value
Operational data	Current memory utilization is {ITEM.LASTVALUE}
Severity	High
Expression	last(/sr-44593-vm01/vm.memory.util)>70
OK event generation	Expression
Problem event generation mode	Single
OK event closes	All problems
Allow manual close	No
URL	(Not set)
Description	The system is running out of free memory.
Status	Enabled

[after adding make a test for it as shown in SS5,6,7

APP 3

Monitoring

Services

Inventory

Reports

Configuration

Host groups

Templates

Hosts

Maintenance

Actions

Discovery

Support

Integrations

Help

User settings

Sign out

Triggers

All hosts / sr-44593-vm01-check description f... Enabled 70% Items 55 Triggers 27 Graphs 10 Discovery rules 3 Web scenarios

Trigger Tags 2 Dependencies

* Name Memory utilization is higher than 70% - call to client

Event name memory utilization is higher than 70% (directly inform to client on call)

Operational data

Severity Not classified Information Warning Average High Disaster

* Expression last(/sr-44593-vm01/vm.memory.util)>70

Expression constructor

OK event generation Expression Recovery expression None

PROBLEM event generation mode Single Multiple

OK event closes All problems All problems if tag values match

Allow manual close

URL

Description The system is running out of free memory.

Enabled

Update Clone Delete Cancel

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The top screenshot shows the 'Triggers' configuration page in Nagios XI. The trigger is named 'Memory utilization is higher than 70% - call to client'. The event name is 'memory utilization is higher than 70% (directly inform to client on call)'. The severity is set to 'High'. The expression is 'A last(/sr-44593-vm01/vm.memory.util)>70'. The target is 'A last(/sr-44593-vm01/vm.memory.util)>70'. The OK event generation is set to 'Expression'. The PROBLEM event generation mode is 'Single'. The OK event closes are 'All problems'. The description is 'The system is running out of free memory.' The trigger is enabled.

The bottom screenshot shows the same configuration page with a 'Test' dialog box open. The 'Test data' section shows 'Expression variable elements' as 'last(/sr-44593-vm01/vm.memory.util)' and 'Result type' as 'Numeric (float)'. The 'Value' field is set to '71'. The 'Result' section shows 'Expression' as 'A last(/sr-44593-vm01/vm.memory.util)>70' and 'Result' as 'TRUE'. The 'Error' field is empty.

Trigger Notes –

1. Memory Utilization

Configuration Summary

- Trigger monitors memory usage of VM
- Host: sr-44593-vm01
- Item used: vm.memory.util
- Threshold: > 70%

Expression Used

last(/sr-44593-vm01/vm.memory.util)>70

Operational Data (Important)

Current memory utilization is {ITEM.LASTVALUE}

✓ This helps show real-time memory value in alerts

Behavior

- Trigger goes to PROBLEM when memory > 70%
- Returns to OK when value goes below threshold
- Event type: Single
- Manual close: Disabled

Disk C: Drive Usage Trigger

Field	Value
Name	C: Drive disk usage is higher than 80%
Event name	C: Drive storage usage critical
Operational data	Current disk usage is {ITEM.LASTVALUE}
Severity	High
Expression	last(/sr-44593-vm01/vfs.fs.size[C:,pused])>80
OK event generation	Expression
Problem event generation mode	Single
OK event closes	All problems
Allow manual close	No
URL	Not set
Description	C drive storage is running low
Status	Enabled

Purpose

Monitors Windows C: drive disk usage and triggers alert when storage exceeds threshold.

Key Configuration

- Metric: vfs.fs.size[C:,pused]
 - Threshold: > 80%
 - Host: sr-44593-vm01
-

Expression

last(/sr-44593-vm01/vfs.fs.size[C:,pused])>80

Operational Data

Current disk usage is {ITEM.LASTVALUE}

✓ Shows real-time disk usage in alerts

Behavior

- PROBLEM → when disk usage > 80%
- OK → when disk usage drops below threshold
- Event type: Single

Memory Utilization Trigger

Trigger Fields

Field	Value
Name	Memory utilization is higher than 70%
Event name	Memory utilization is higher than 70% (call to client)
Operational data	Current memory utilization is {ITEM.LASTVALUE}
Severity	High
Expression	last(/sr-44593-vm01/vm.memory.util)>70
OK event generation	Expression
Problem event generation mode	Single
OK event closes	All problems
Allow manual close	No
URL	Not set
Description	The system is running out of free memory
Status	Enabled

Expression Used

```
last(/sr-44593-vm01/vm.memory.util)>70
```

Operational Data Purpose

- Shows real-time memory usage in problem alerts
- Helps support team quickly understand severity

Example output in alert:

Current memory utilization is 82%

Behavior

- Trigger = PROBLEM when memory > 70%
- Trigger = OK when memory < 70%
- Event type = Single

We now configure **alerting system (Users + Media + Actions)**

1. Create User Group

Navigation

```
Administration
├── User groups
│   └── Create user group
```

Fields

Field	Value
Group name	Cahal computers/Jeena sikho / Custom

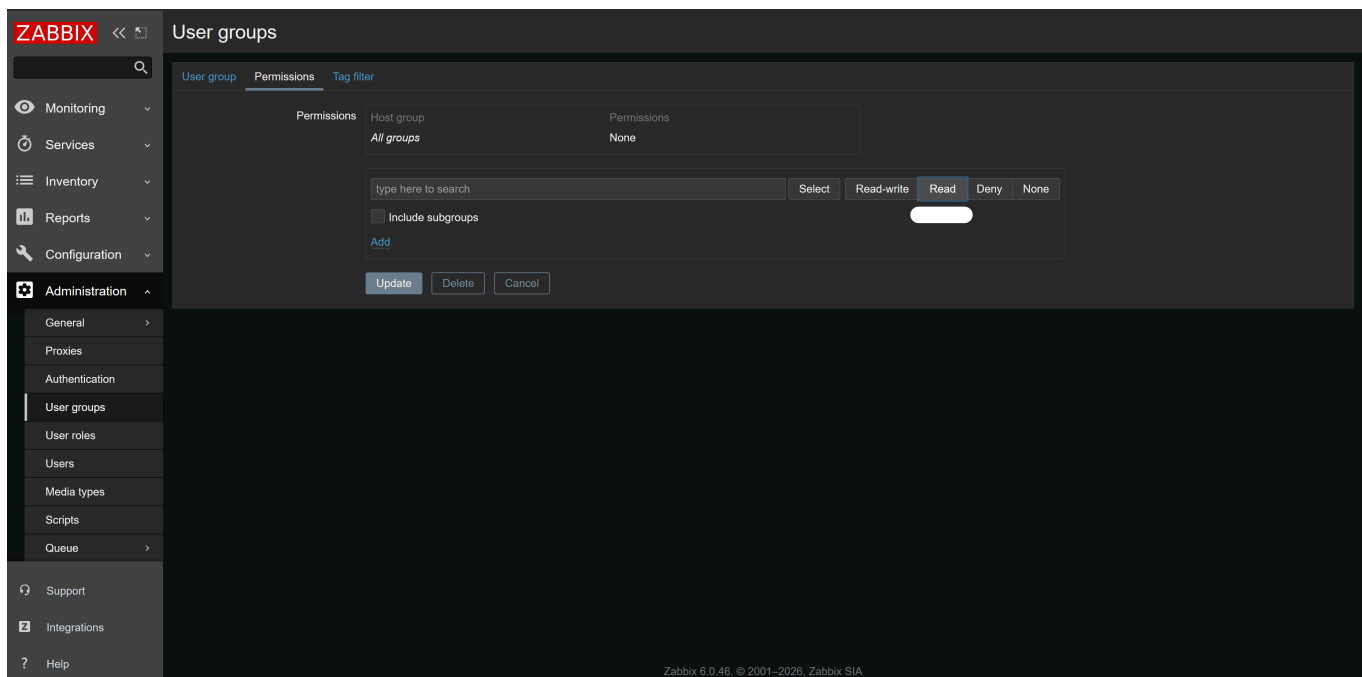
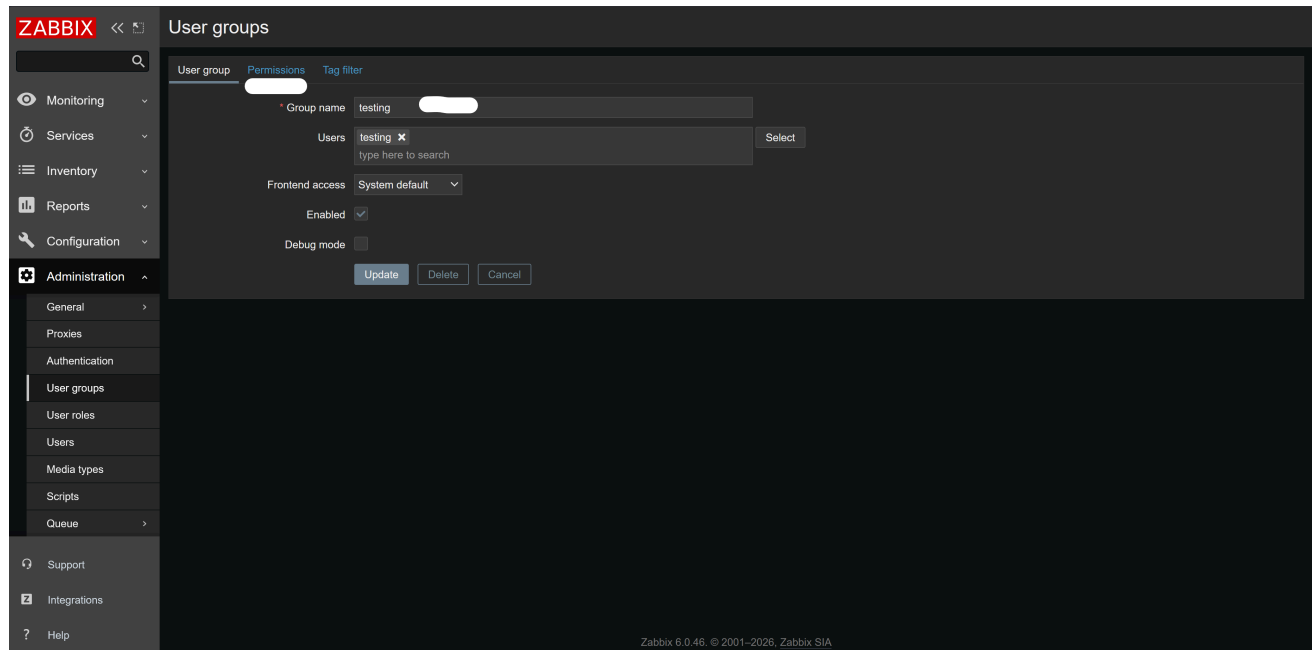
Permissions Tab

Field	Value
Host group	Linux Servers
Permission level	Read

Purpose

- Controls which hosts users can see
- Required for receiving alerts

Shown in SS8,9



2. Create User Navigation

```
Administration
└─ Users
    └─ Create user
```

User Tab

Field	Value
Username	zabbix-admin / noc-user/Custom
Name	Optional
Surname	Optional
Password	Strong password

Groups Tab

- Assign user to:
 - Linux Admins / NOC Team / Custom

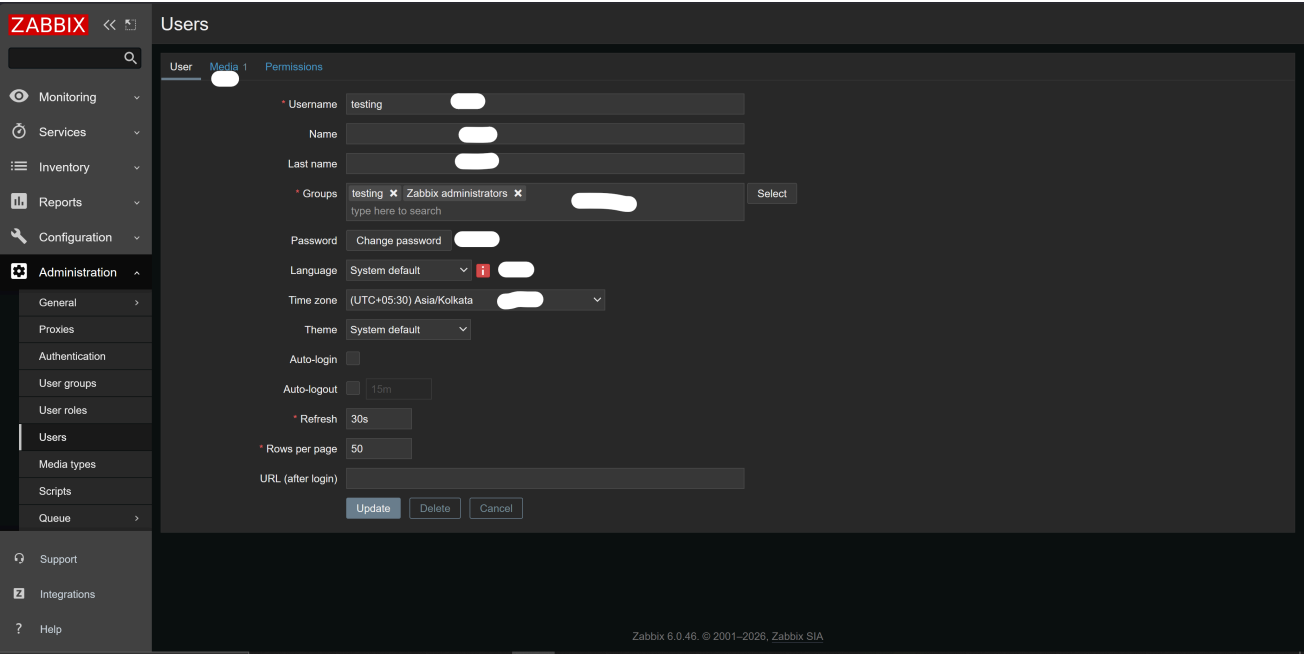
Role Tab

Field	Value
Role	Always CLIENT ROLE User (or Admin if required)

Purpose

- Receives alerts
- Linked with Media (Email)

Shown in SS10,11,12



ZABBIX

Monitoring

Services

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Integrations

Help

Users

UserMedia 1Permissions

Media	Type	Send to	When active	Use if severity	Status	Action
Zabbix Server		akrac9927@gmail.com	1-7,00:00-24:00	N I W A H D	Enabled	Edit Remove

Add

UpdateDeleteCancel

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Users

UserMedia 1Permissions

RoleSuper admin role XSelect

User typeSuper admin

Permissions

Host groupAll groups

PermissionsRead-write

Permissions can be assigned for user groups only.

Access to UI elements

MonitoringDashboardProblemsHostsLatest dataMapsDiscovery

ServicesServicesService actionsSLASLA report

InventoryOverviewHosts

ReportsSystem informationScheduled reportsAvailability reportTriggers top 100AuditAction logNotifications

ConfigurationHost groupsTemplatesHostsMaintenanceActionsEvent correlationDiscovery

AdministrationGeneralProxiesAuthenticationUser groupsUser rolesUsersMedia typesScriptsQueue

Access to services

Read-write access to servicesAll

Read-only access to servicesAll

Access to modules

No enabled modules found.

Access to API

Enabled

3. Configure Media Type (Email)

Navigation

Alerts

- Media types
 - Create media type

Email Media Type

Fields

Field	Value
Name	Email SMTP
Type	Email

SMTP Settings

Field	Value
SMTP server	smtp.gmail.com
Port	587
Security	STARTTLS
Username	yourmail@gmail.com
Password	App Password

Purpose

- Defines HOW alerts are sent (Email)

Shown in SS13,14,S2

Media types

Media type Message templates 6 Options

* Name

SAKAAR CORPORATE SERVICES

Type

Email

* SMTP server

zimbra187-ind.megavelocity.net

SMTP server port

465

* SMTP helo

zimbra187-ind.megavelocity.net

* SMTP email

sakaarsmtp@zimbra187-ind.megavelocity.net

Connection security

None

STARTTLS

SSL/TLS

SSL verify peer

☐

SSL verify host

☒

Authentication

None

Username and password

Username

sakaarsmtp@zimbra187-i

Password

Change password

Message format

HTML

Plain text

Description

Bgh(ew863dWP6d&f

Enabled

☒

Update

Clone

Delete

Cancel

ZABBIX << Media types

Media type Message templates 5 Options

* Name

Zabbix Server

Type

Email

* SMTP server

smtp.gmail.com

SMTP server port

587

* SMTP helo

gmail.com

* SMTP email

akrao9927@gmail.com

Connection security

None

STARTTLS

SSL/TLS

SSL verify peer

☐

SSL verify host

☐

Authentication

None

Username and password

Username

akrao9927@gmail.com

Password

Change password

Message format

HTML

Plain text

Description

ynmdlpxfdmfsa

Enabled

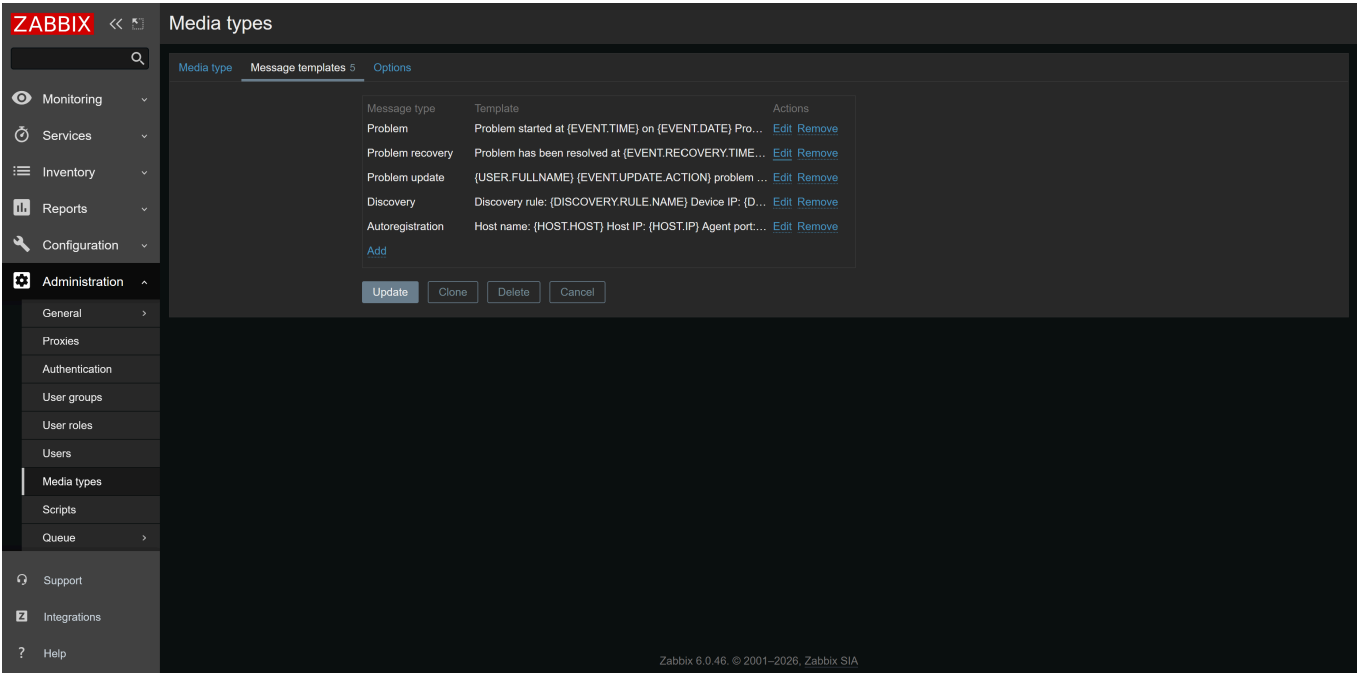
☒

Update

Clone

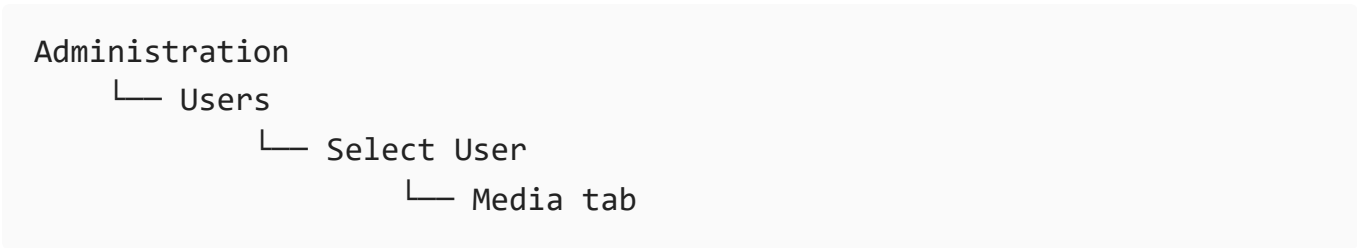
Delete

Cancel



4. Link Media to User

Navigation



Media Tab Fields

Field	Value
Type	Email SMTP / Telegram
Send to	email@example.com
Active period	1-7, 00:00-24:00

Severity Selection

Enable all required:

- Not classified
- Information
- Warning
- Average
- High
- Disaster

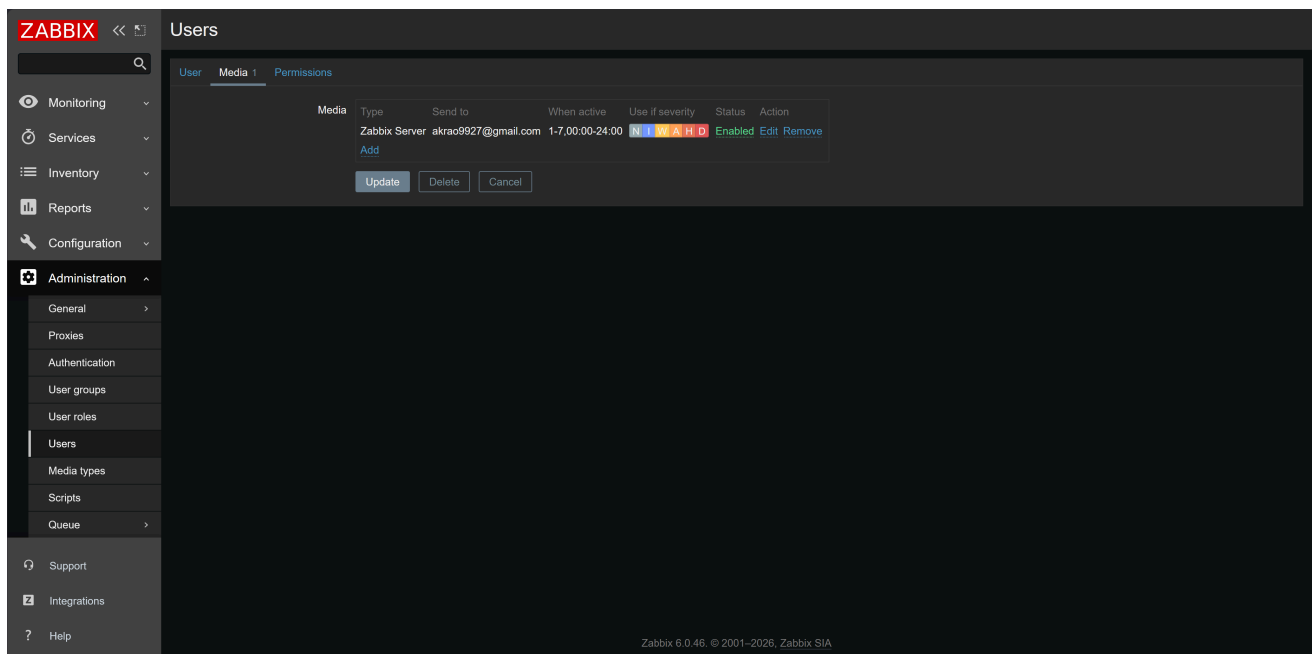
Purpose

- Connects USER → NOTIFICATION CHANNEL

Without this step:

✗ Alerts will NOT be sent

Shown in SS11

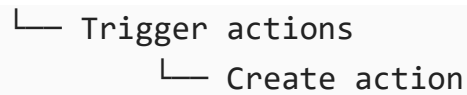


5. Trigger Action

A Trigger Action is an automation rule that tells Zabbix what to do when a trigger generates a Problem or Recovery event.

Navigation

Alerts
└─ Actions



Action Components

1. Action Name

Provide a meaningful name that describes the purpose of the action.

Examples:

- Linux Server Email Alerts
- Windows Server Notifications
- Production Critical Alerts
- Zimbra Monitoring Alerts

2. Conditions

Conditions determine when the action should execute.

Common conditions include:

Condition	Example
Event Source	Trigger
Host Group	Linux Servers
Severity	High
Trigger Name	Contains "Memory"
Tag	Environment = Production

The action runs only when all configured conditions are satisfied.

3. Operations

Operations define what should happen when a problem event occurs.

Common operations:

- Send Email
- Send Telegram Message
- Send Slack Notification
- Execute Remote Command
- Call a Webhook

Example:

Operation:

Send Message

Recipient:

Linux Admins

Media Type:

Email SMTP

4. Recovery Operations

Recovery Operations define what happens after the trigger returns to the OK state.

Example:

Send Recovery Email

Recipient:

Linux Admins

This informs administrators that the issue has been resolved.

5. Update Operations (Optional)

Update Operations are executed when an event is updated, such as:

- Problem acknowledged
- Severity changed
- Tags updated

Example:

Notify Team Lead when the problem is acknowledged.

Example Configuration

Action Name

Linux Server Email Alerts

Conditions

Event Source = Trigger

Host Group = Linux Servers

Severity >= High

Operations

Send Email

To:

Linux Admins

Media:

Email SMTP

Recovery Operations

Send Recovery Email

To:

Linux Admins

Shown in SS15,16,17

ZABBIX

<<

>>

Monitoring

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Actions

Operations 2

Name

zabbix server

Conditions

Label

Name

Action

Add

Enabled

☒

At least one operation must exist.

Update

Clone

Delete

Cancel

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ZABBIX

<<

>>

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Sign out

Actions

Operations 2

Default operation step duration

1h

Operations

Steps

Details

Start in

Duration

Action

1

Send message to users: zabbix Server (Zabbix Server) via Zabbix Server

Immediately

Default

Edit

Remove

Send message to user groups: Zabbix administrators via Zabbix Server

Add

Recovery operations

Details

Send message to users: zabbix Server (Zabbix Server) via Zabbix Server

Send message to user groups: Zabbix administrators via Zabbix Server

Add

Edit

Remove

Update operations

Details

Action

Add

Pause operations for suppressed problems

☒

Notify about canceled escalations

☒

At least one operation must exist.

Update

Clone

Delete

Cancel

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ZABBIX

Monitoring

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Actions

Action

Operations 2

Default operation step duration

Operations

Recovery operations

Update operations

Pause operations for suppressed problems

Notify about canceled escalations

* At least one user or user group must be selected.

Update

Operation details

Operation

Send message

Steps

1

1

(0 - infinitely)

Step duration

0

(0 - use action default)

* At least one user or user group must be selected.

Send to user groups

User group

Zabbix administrators

Action

Remove

Add

Send to users

User

zabbix Server (Zabbix Server)

Action

Remove

Add

Send only to

Zabbix Server

Custom message

Subject

[PROBLEM] [HOST.NAME]

Message

<div style="font-family: Arial; font-size: 13px; color: #333;">
<div style="font-weight: bold; font-size: 16px; margin-bottom: 10px;">
Zabbix Monitoring Alert
</div>
</div>

Update

Cancel

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